Concepts of Biostatistics and Data Science

A two-day crash course with practical analysis in R

Day 2: Regression and an introduction to data science



Plan for today

Time	Session	Focus
09:00 to 09:10	Theory 1	Recap; four purposes of analysis
09:10 to 09:25	Theory 2	Scatterplots, correlation, association and causation
09:25 to 09:45	Theory 3	Simple linear regression
09:45 to 10:10	Theory 4	Multiple regression, confounding and mediators
10:10 to 10:20	Theory 5	Model checks and plots of fitted relationships
10:20 to 10:40	Theory 6	Introduction to logistic and Poisson regression
10:40 to 11:00	Theory 7	Prediction, train/test, and clustering
11:00 to 11:30	Break	(not counted in teaching time)
11:30 to 13:30	Practical	Regression in R, guided demos, final exercise

By the end of today you will be able to

- Explain the difference between describing, estimating associations, predicting and discovering groups
- Use a scatterplot and correlation to describe a relationship, and explain why association is not causation
- Fit and interpret simple and multiple linear regression models in R
- Explain confounding and mediation, and interpret an adjusted association
- Check a linear regression model using residual plots, and plot fitted relationships
- Read the output of a logistic and a Poisson regression (OR, aOR, IRR) at an introductory level
- Describe what k-means clustering does, why scaling matters, and why clusters need checking

Recap of Day 1

- Start from the question, the outcome type and the design (independent or paired)
- Describe first: counts with denominators, mean (SD) or median (IQR), plots
- Report an effect size with a 95% CI; a p-value alone is not enough
- Welch's t-test and ANOVA by default; rank-based tests answer a different question
- Proportions, risks and odds differ; the design decides which you can interpret
- Crude associations can be confounded: smoking OR 3.4 fell to 2.6 after stratifying by age

Four purposes of analysis

Purpose	Example question (our data)	Typical tools	Judged by
Describe	What is the distribution of SBP?	Summaries, plots	Accuracy, clarity
Estimate associations	Is activity associated with lower SBP, at the same age and sex?	Regression with adjustment	Bias, CI, design
Predict	Can we predict a person's SBP from routine data?	Regression, machine learning	Error on new data
Discover groups	Are there distinct lifestyle profiles?	Clustering	Usefulness, stability

> The same model can serve different purposes, but each purpose asks different things of it.

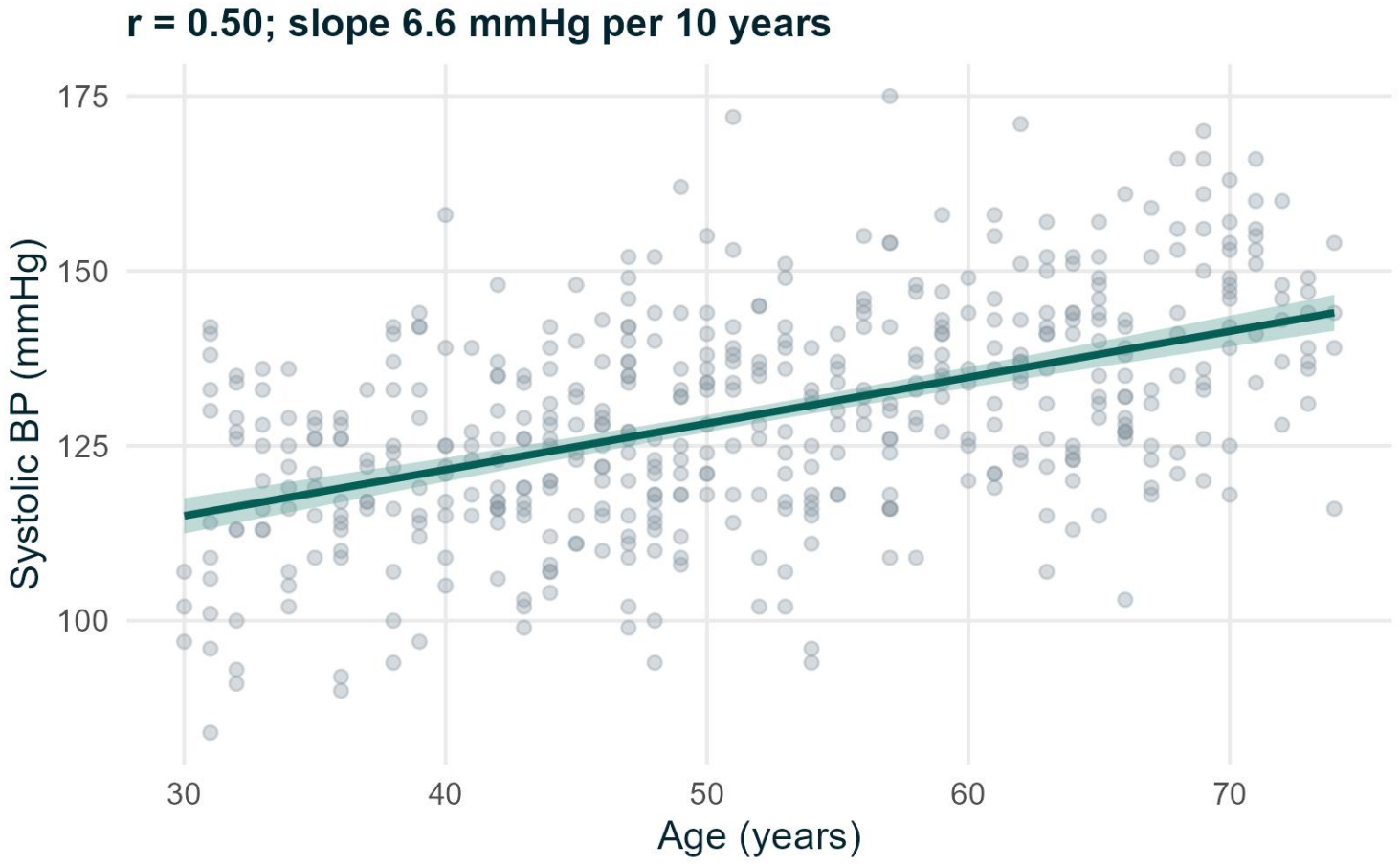


Relationships between continuous variables

Part 1. Scatterplots, correlation, association and causation

Always start with a scatterplot

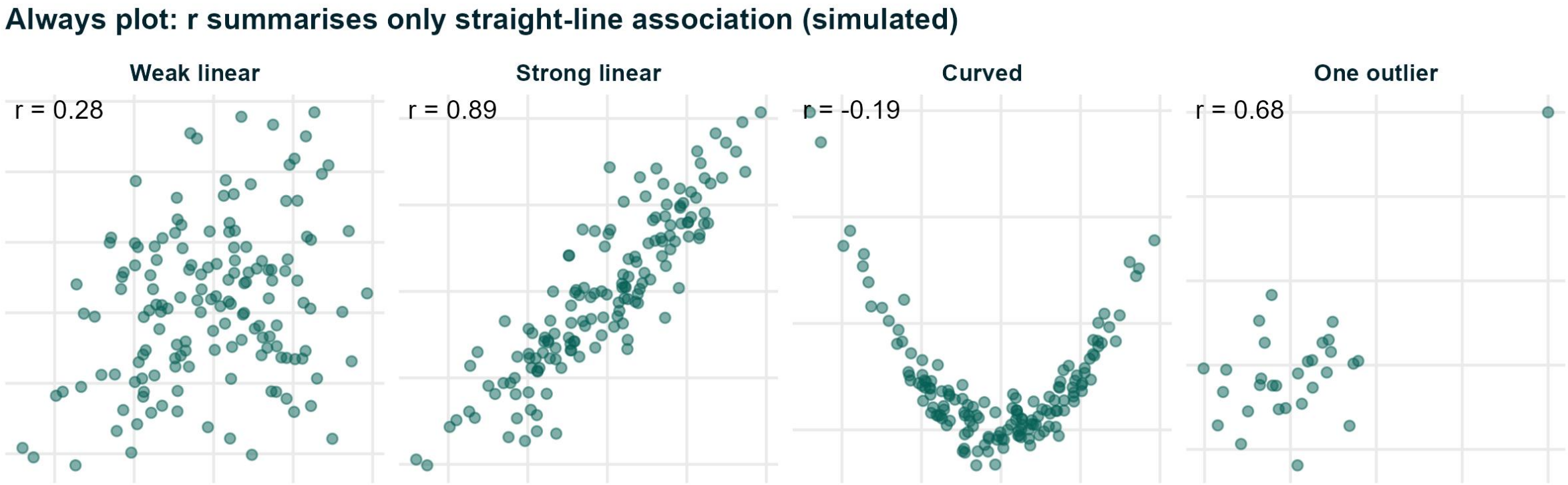
- Each point is one person
- Look for direction, form (straight or curved), spread and outliers
- SBP rises with age, fairly linearly
- Line: least-squares fit with 95% confidence band



Correlation coefficients

- Pearson's r: strength of a straight-line association, from -1 to +1
  - Age and SBP: r = 0.50 (95% CI 0.43 to 0.57)
- Spearman's rho: uses ranks; measures a monotonic association, less affected by skew and outliers
  - Activity and CRP: rho = -0.15, p = 0.001
- r has no units and does not tell you how much y changes per unit of x
- r near 0 does not mean no relationship: it may be curved

Why we always plot



Simulated examples. The curved relationship is strong but r is close to 0; one outlier creates r = 0.68.

Association is not causation

- An association between X and Y can arise because:
  - X causes Y
  - Y causes X (reverse causation)
  - A third variable causes both (confounding)
  - Chance, or how participants were selected or measured (bias)
- More active people have lower SBP in our data, but they are also younger
- Randomised trials remove confounding by design; observational studies must adjust for it



Simple linear regression

Part 2. One outcome, one predictor

The regression line

- Model: SBP = b0 + b1 × age + error
- b1 (slope): difference in mean SBP for a 1-unit difference in age
- b0 (intercept): predicted mean SBP when age = 0 (anchors the line; often not meaningful)
- Error: how far each person is from the line (the residual)
- R chooses b0 and b1 to make the squared residuals as small as possible (least squares)

```
R:  fit1 <- lm(sbp_0 ~ age, data = hearts);  summary(fit1);  confint(fit1)
```

Reading the output

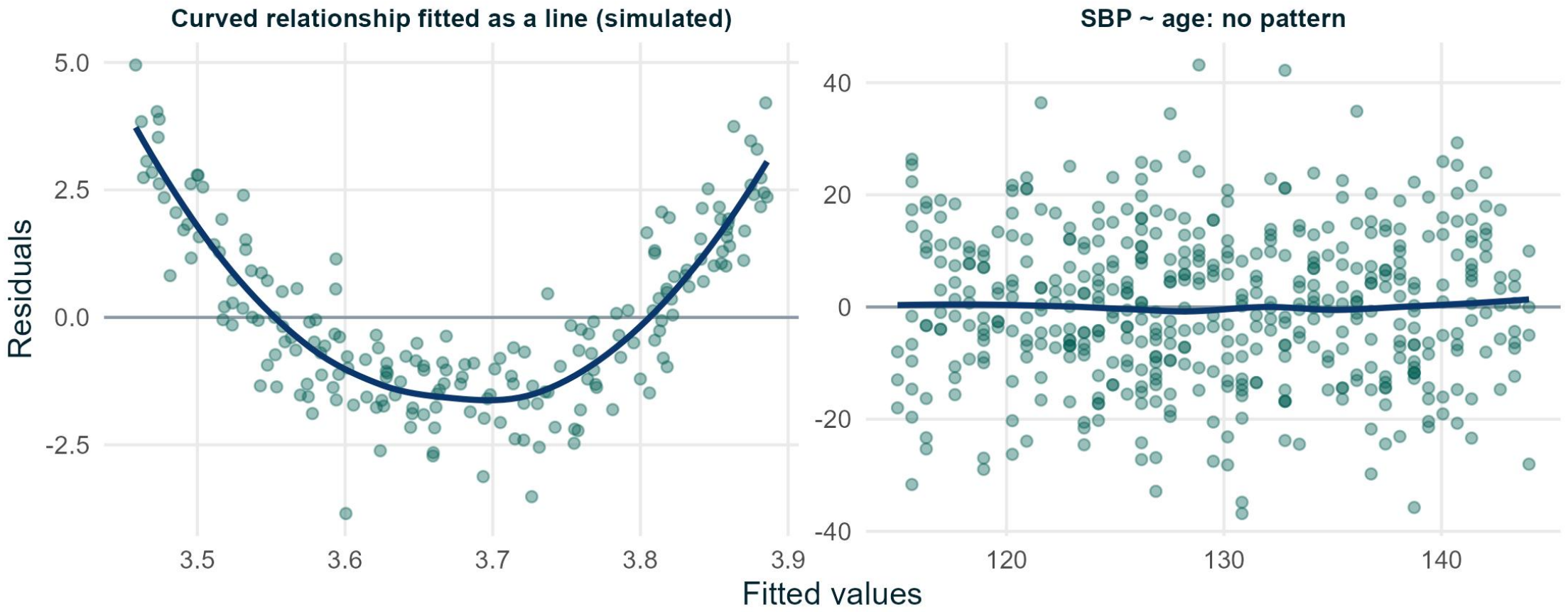
Term	Estimate	95% CI	Meaning
Intercept	95.2	89.8 to 100.6	Predicted mean SBP at age 0 (outside the data)
age	0.66	0.56 to 0.76	Mean SBP 0.66 mmHg higher per year of age
age10 (per 10 years)	6.6	5.6 to 7.6	Mean SBP 6.6 mmHg higher per decade

- $R^2 = 0.25$: age accounts for a quarter of the variation in SBP
- Residual standard error 13.5 mmHg: typical distance of a person from the line
- Rescale predictors (e.g. per 10 years) so the slope is a meaningful difference

Assumptions of linear regression

- Independent observations: comes from the design, not from a plot
- Linearity: the mean outcome changes in a straight line with each predictor
- Constant spread: residuals have similar variance across fitted values
- Approximately normal residuals: matters least with large samples
- The OUTCOME does not need to be normally distributed; the residuals are what we check
- Watch for single points with large influence on the fit

Checking residuals



Left: SBP on age, no pattern. Right: a curved relationship forced into a straight line leaves a clear pattern.

Categorical predictors: t-tests and ANOVA are regressions

- A binary predictor gives the difference in means: same as a Student t-test
- A predictor with k categories becomes k - 1 indicator (dummy) variables
  - Each coefficient is a difference from the reference level
- SBP by district, adjusted for age and sex (reference North):
  - South +5.7 mmHg (2.3 to 9.1); East +2.0 (-1.3 to 5.3); West -1.6 (-4.9 to 1.7)
- Regression lets us compare groups while adjusting for other variables



Multiple regression and confounding

Part 3. Adjusted associations, confounders and mediators

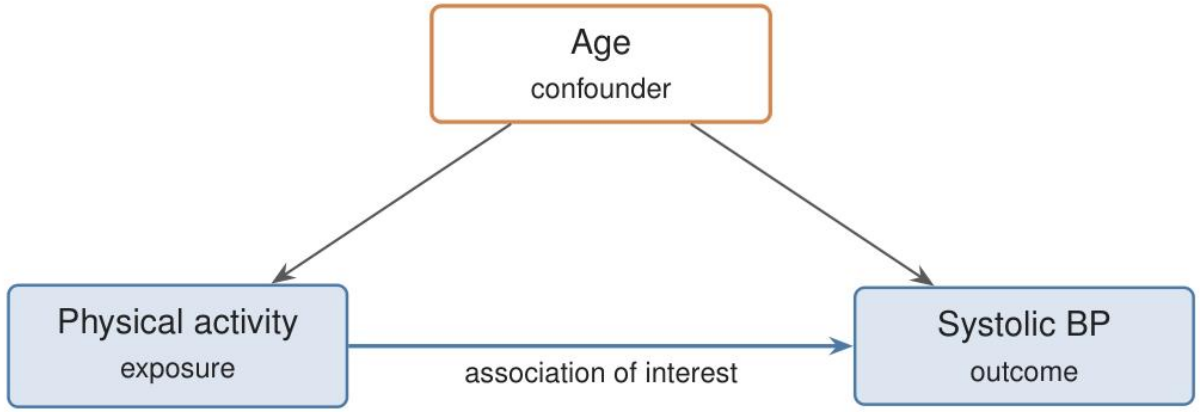
Why adjust?

- Question: is physical activity associated with lower SBP?
- Crude: each extra hour of activity per week, SBP 2.1 mmHg lower
- But older people are less active (r = -0.19) and have higher SBP
- Part of the crude association may reflect age
- Multiple regression compares people with the same values of the other variables

> Model: SBP = b0 + b1 × activity + b2 × age + b3 × male + error

Confounder: a common cause

- Age affects activity and affects SBP
- It is not on the path from activity to SBP
- Ignoring it mixes the effect of age into the activity estimate
- Adjust for confounders to estimate the activity association

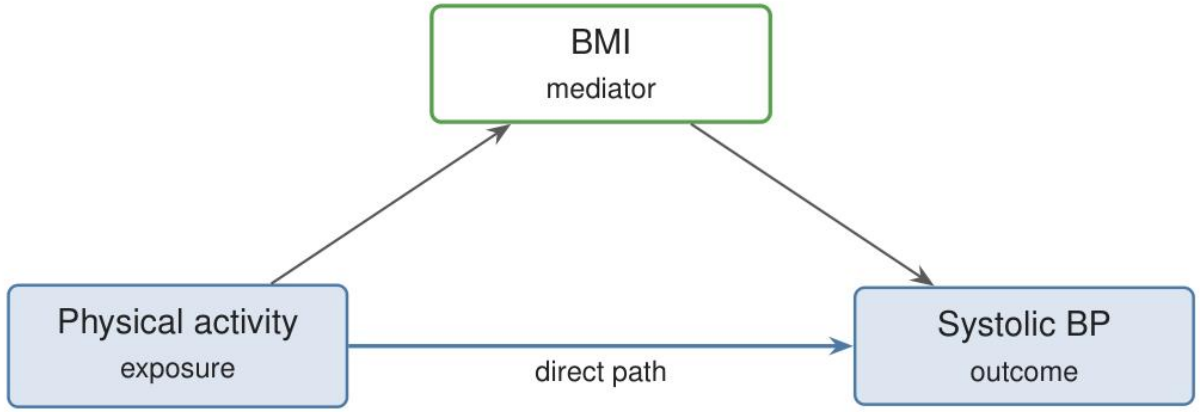


Older people are less active and have higher SBP. Part of the crude association between activity and SBP is due to age. **Adjust for a confounder.**

Course data: –2.08 mmHg per hour/week crude; –1.60 adjusted for age and sex.

Mediator: on the causal path

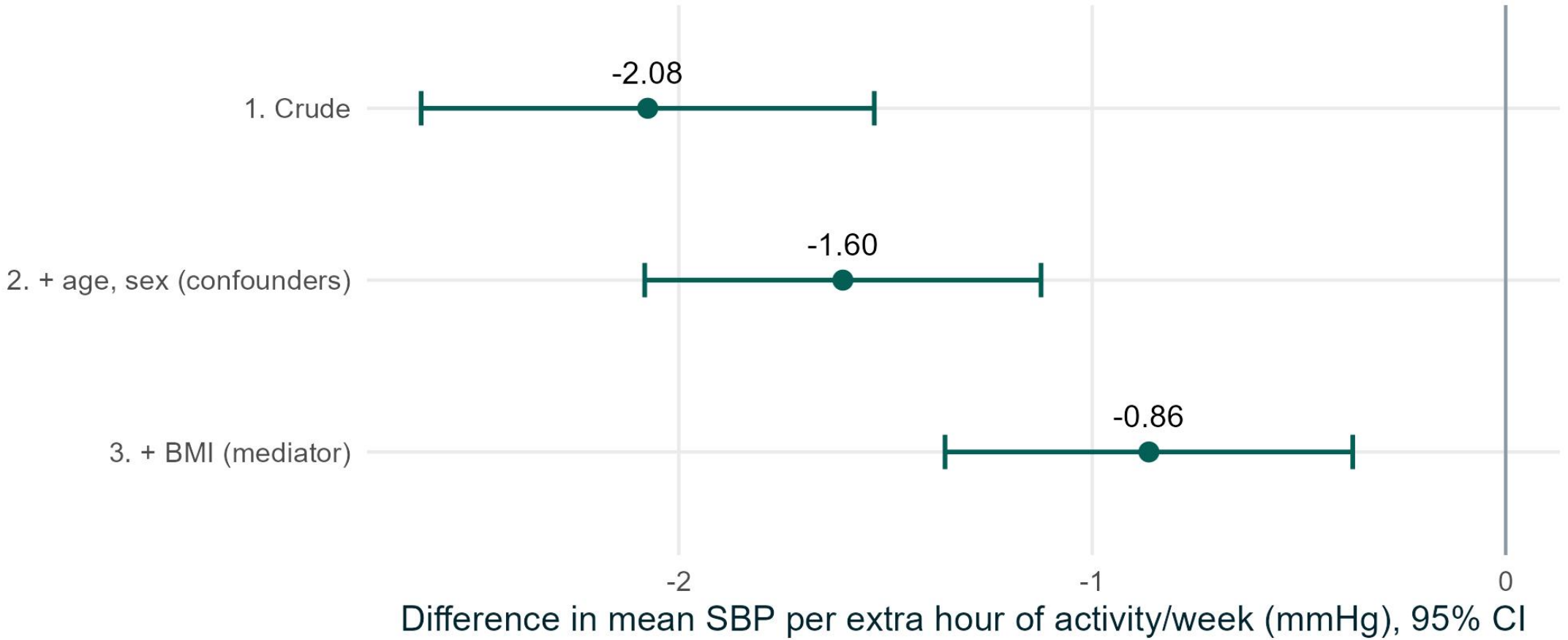
- Activity may lower BMI, and BMI raises SBP
- BMI carries part of the effect of activity
- Adjusting for BMI removes that part
- The result is a different quantity: the association not through BMI



Activity can lower BMI, and BMI raises SBP: BMI lies on the path. Adjusting for BMI removes the part of the effect that works through BMI. **Do not adjust for a mediator** when you want the total effect.

Course data: −1.60 mmHg per hour/week (age, sex); −0.86 when BMI is also added.

Crude, adjusted and mediator-adjusted estimates



Activity coefficient (mmHg per hour/week) with 95% CI. Models 1 and 2 use 473 people, model 3 uses 463.

Choosing what to adjust for

- Use subject knowledge and a causal diagram, not p-values or automatic stepwise selection
- Adjust for confounders: common causes of exposure and outcome
- Do not adjust for mediators if you want the total effect
- Do not adjust for consequences of the outcome, or for colliders (common effects)
- Keep enough data: as a rough guide, at least 10 to 20 observations per coefficient
- Adjustment cannot fix unmeasured or poorly measured confounders

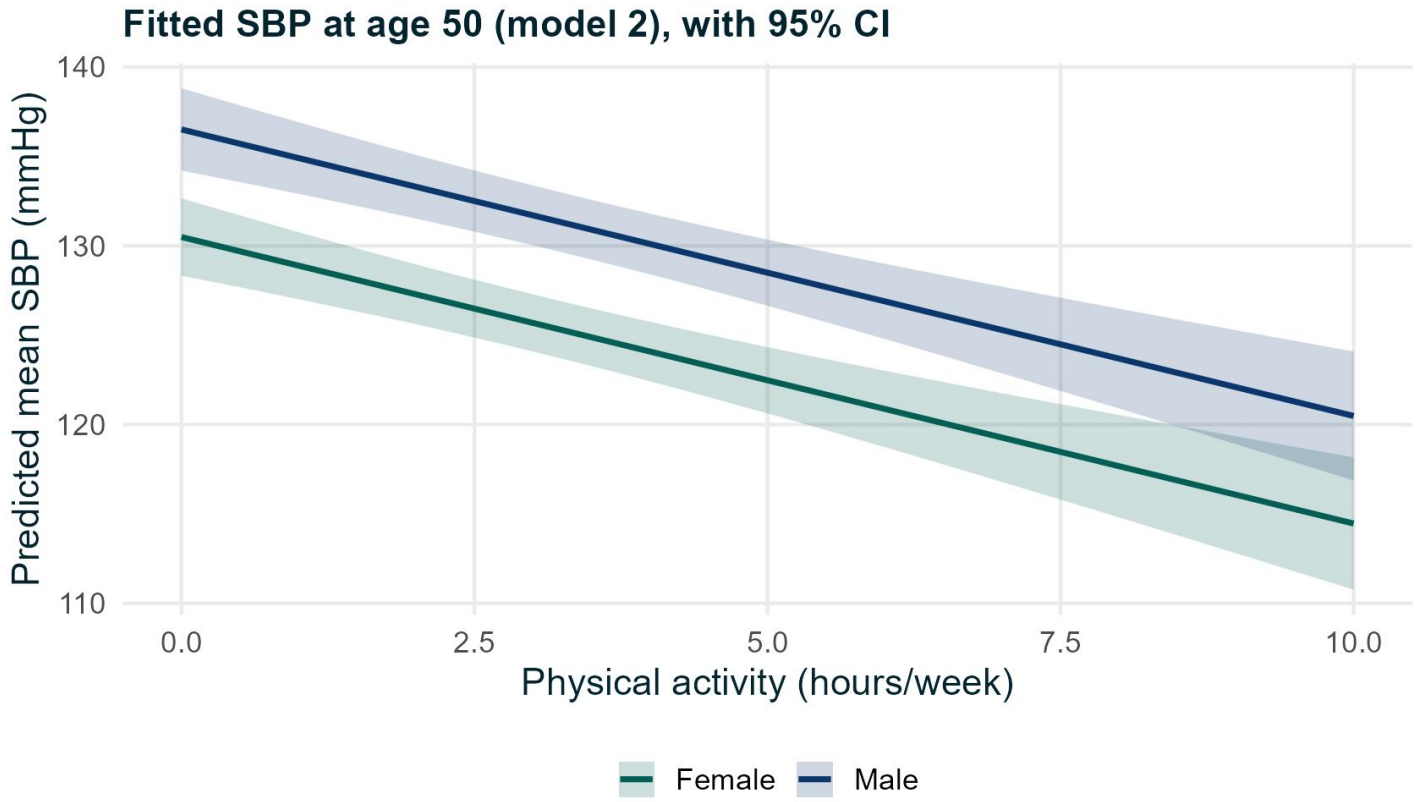
Interpreting a multiple regression

Term	Estimate	95% CI	Interpretation
activity_hr	-1.60	-2.08 to -1.12	Per extra hour/week, same age and sex
age	0.59	0.49 to 0.69	Per year of age, same activity and sex
sexMale	6.0	3.7 to 8.3	Men vs women, same age and activity

- Each coefficient is adjusted for the others in the model
- Adjusted R^2 = 0.34; 7 people excluded because activity was missing
- These are associations: causal interpretation depends on design and assumptions

Plotting a fitted relationship

- Choose values for the other variables (here age 50)
- Predict the mean outcome over a range of the exposure
- Show the 95% confidence band
- Parallel lines: the model assumes the same activity slope for both sexes



Effect modification (interaction): different from confounding

- Confounding: a third variable distorts the overall association. Remedy: adjust
- Effect modification: the association genuinely differs between subgroups
  - Example question: is the smoking-BP association stronger in men?
- Fit an interaction term, e.g. lm(sbp_0 ~ smoker * sex + age, data = hearts)
- Interactions need much larger samples to estimate well; pre-specify them



Beyond continuous outcomes

Part 4. An introduction to logistic and Poisson regression

Matching the model to the outcome

Outcome	Model	Effect measure	In this course
Continuous	Linear regression	Difference in means	Practised
Binary	Logistic regression	Odds ratio (OR, aOR)	Introduced, guided demo
Count, with exposure time	Poisson regression (offset)	Incidence rate ratio (IRR)	Introduced, guided demo
Ordinal	Ordinal (proportional odds) regression	Cumulative OR	Further learning
Nominal, > 2 categories	Multinomial regression	Relative risk ratios	Further learning
Time-to-event	Cox regression, Kaplan-Meier	Hazard ratio	Further learning
Repeated / clustered	Mixed models, GEE	Depends on outcome	Further learning

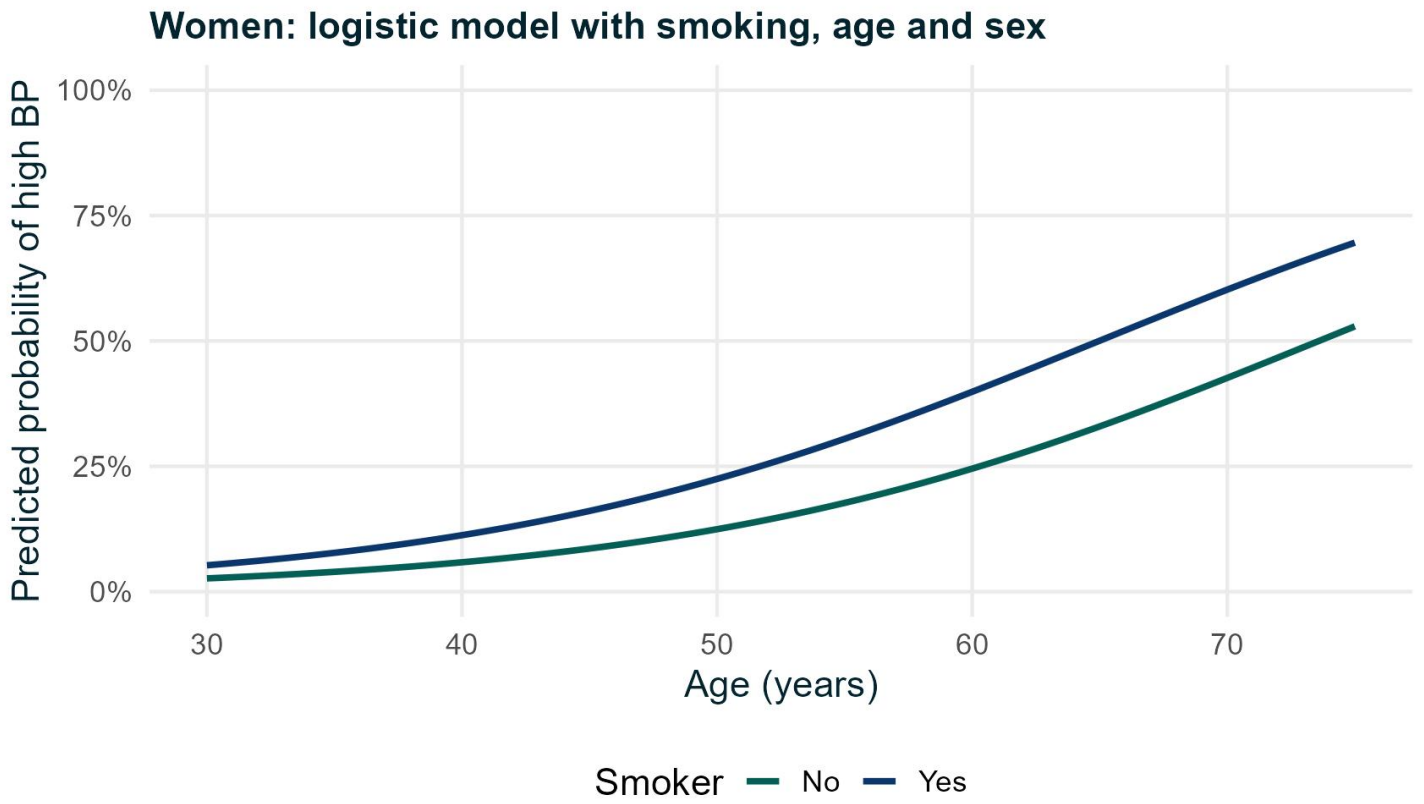
Logistic regression for a binary outcome

- Models the log-odds of the outcome as a linear function of the predictors
- exp(coefficient) = odds ratio; with other variables in the model, an adjusted OR (aOR)
- High BP and smoking: crude OR 3.39 (2.05 to 5.61)
- Adjusted for age and sex: aOR 2.04 (95% CI 1.17 to 3.54)
- Age: aOR 2.29 per 10 years (1.84 to 2.87); men vs women: aOR 2.06 (1.30 to 3.30)

```
glm(high_bp_0 ~ smoker + age10 + sex, family = binomial, data = hearts)  then  tidy(fit, conf.int = TRUE, exponentiate = TRUE)
```

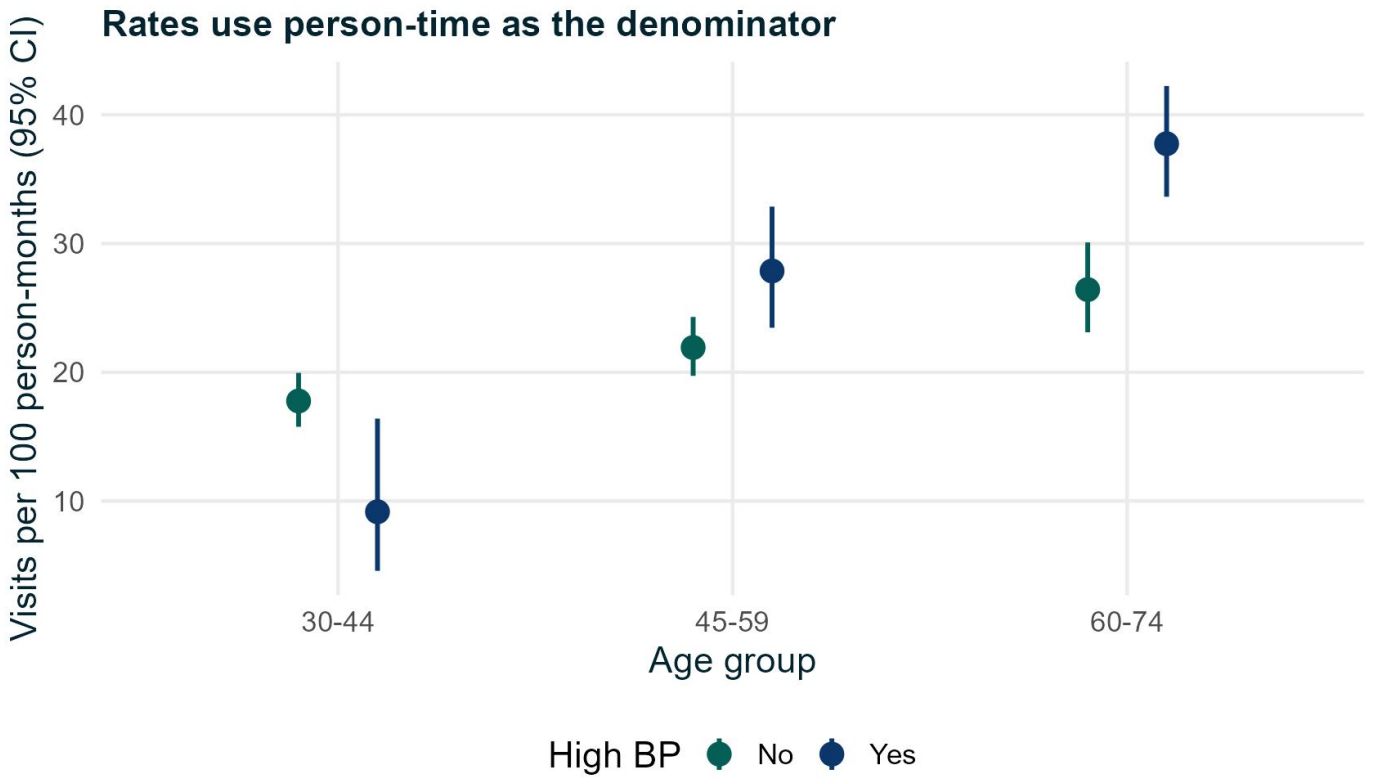
From odds ratios to probabilities

- Predicted probabilities are easier to communicate
- At 60 years, a woman who smokes has a higher predicted probability than a non-smoker
- The OR is constant; the difference in probability is not
- With a common outcome, OR overstates the risk ratio



Poisson regression for counts and rates

- Outcome: number of unplanned clinic visits
- People were followed for 2 to 12 months
- Compare rates: visits per person-month
- Offset log(follow-up months) puts exposure time in the model
- exp(coefficient) = incidence rate ratio (IRR)



Rate ratios and overdispersion

Model	IRR: programme vs standard	95% CI
Poisson	0.91	0.82 to 1.02
Quasi-Poisson	0.91	0.78 to 1.06
Negative binomial (extension)	0.93	0.79 to 1.08

- Poisson assumes variance = mean. Here the dispersion ratio is 2.0
- Overdispersion makes Poisson CIs too narrow; quasi-Poisson or negative binomial widen them
- Adjusted for age and baseline high BP. Conclusion: no clear evidence that visit rates differ



Prediction and discovering groups

Part 5. Train/test separation, leakage and clustering

Explaining versus predicting

Explaining (estimation)

- Goal: estimate an association, ideally a causal one
- Choose variables from a causal diagram
- Report effect sizes with CIs
- Bias matters more than accuracy

Predicting

- Goal: accurate predictions for new people
- Variables chosen for predictive value, not causal role
- Report prediction error on new data
- Coefficients need not have a causal meaning

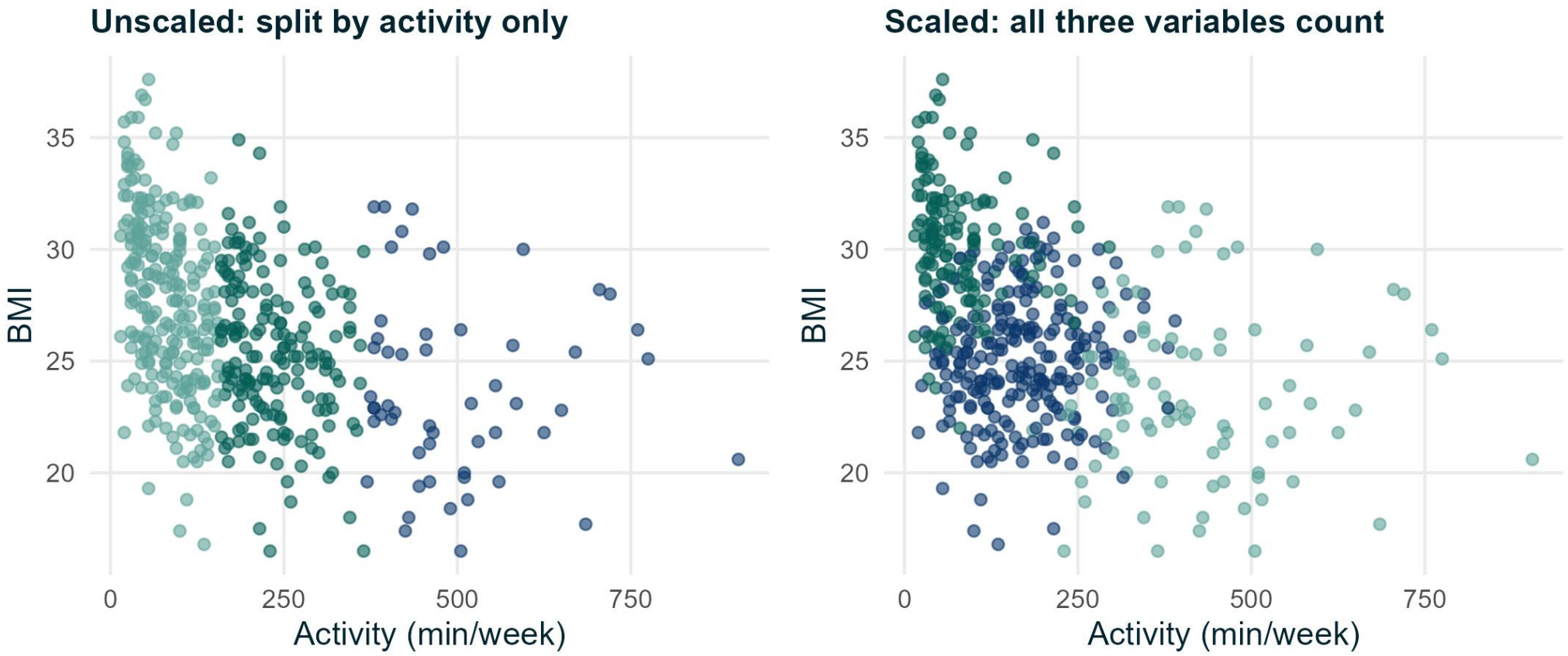
Train/test separation and data leakage

- A model always fits the data it was built on better than new data
- Split the data: fit on a training set (e.g. 70%), evaluate once on a test set (30%)
- Example: predicting SBP. Error (RMSE) in mmHg:
  - Age only: 13.4 training, 13.7 test
  - Age, sex, BMI, activity, smoking, district: 11.4 training, 12.3 test
- Leakage: information from the test set or the future enters model building
  - Choosing variables using all the data; using 12-month SBP to predict 12-month high BP

Discovering groups: clustering

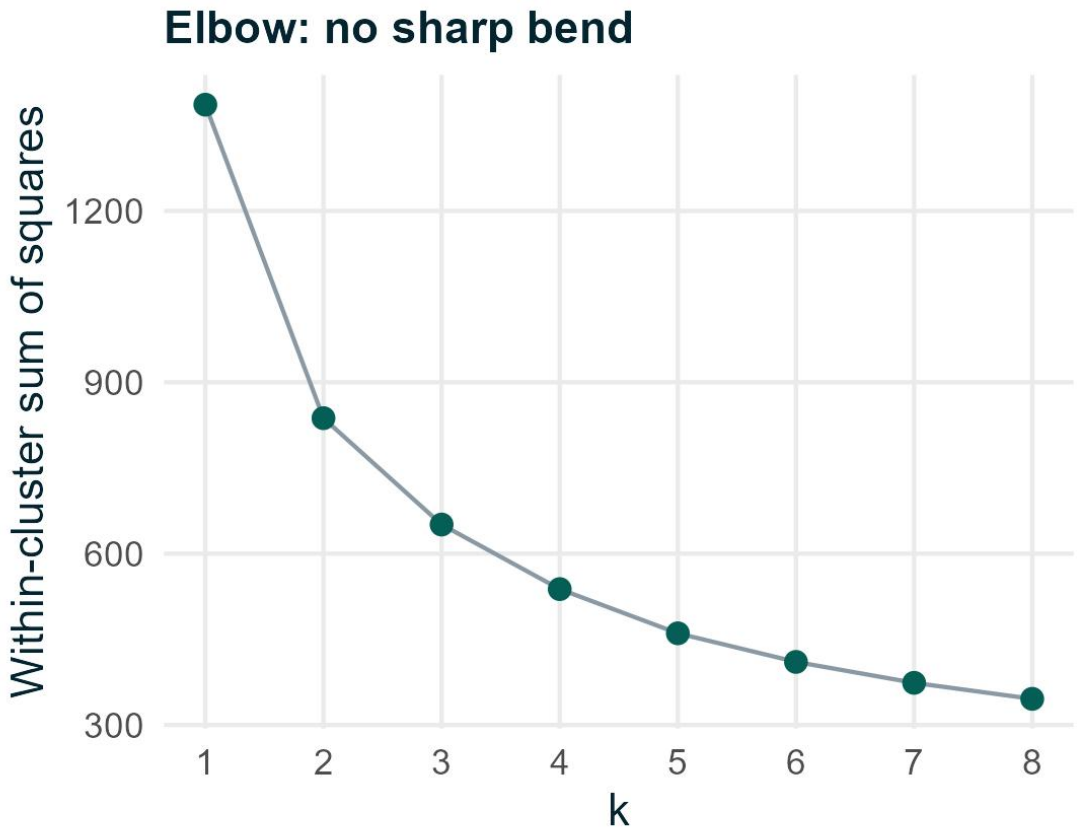
- Unsupervised learning: no outcome variable
- Groups people who are similar on the variables we choose
- Similarity is measured by a distance (usually Euclidean)
- k-means: choose k; assign each person to the nearest centre; move centres to the mean; repeat
- Hierarchical: merge the closest groups step by step, giving a tree (dendrogram)
- Our example: activity, diet score and BMI

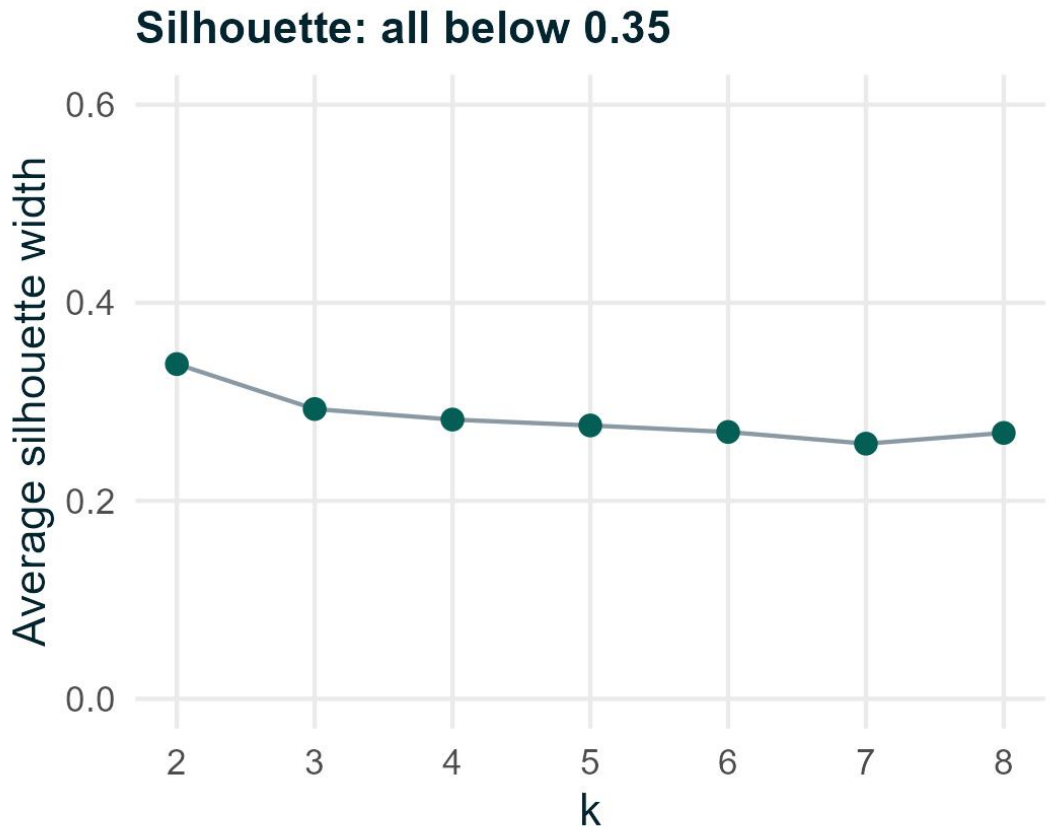
Scaling matters



Activity is measured in hundreds of minutes, diet on a 0 to 10 scale. Without scaling, activity dominates the distances.

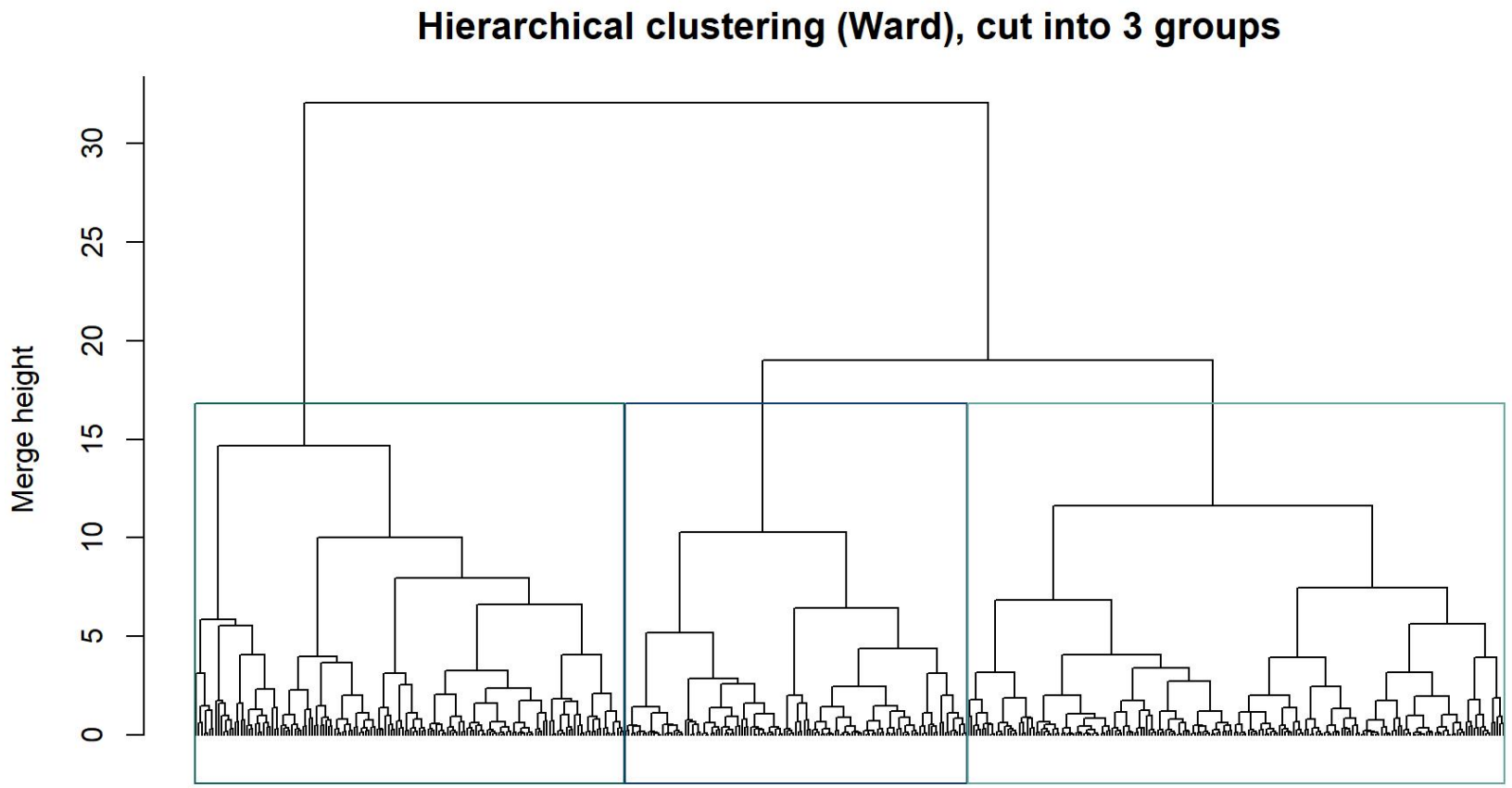
How many clusters?





No sharp elbow; average silhouette widths are all below 0.35. The data do not clearly favour one k.

Hierarchical clustering



Ward's method on the same scaled variables. Cutting the tree at three groups gives clusters similar to k-means.

Are the clusters meaningful?

- Clusters depend on the variables, the scaling, k and the algorithm
- Describe clusters on the original scale and name them carefully
  - Our k = 3: low activity / higher BMI; middle; high activity / good diet / lower BMI
- Check stability: different seeds, subsets, methods
- Check usefulness: do clusters differ on variables NOT used to build them? (mean SBP 138, 127, 121 mmHg)
- Do not test for differences on the clustering variables: clusters were built to differ on them

Where to go next

If your data have...	Look into	Starting point in R
Ordered categories as outcome	Ordinal (proportional odds) regression	MASS::polr()
Unordered categories (> 2)	Multinomial regression	nnet::multinom()
Time to an event, with censoring	Kaplan-Meier, Cox regression	survival package
Repeated measures or clusters	Mixed models, GEE	lme4, geepack
Causal questions from observational data	Causal diagrams, propensity scores	dagitty, MatchIt

Practical session (2 hours)

Part	Script	Workbook	Time
A. Correlation, simple and multiple regression, checks	05_correlation_regression.R	2.1 to 2.4	50 min
B. Logistic and Poisson regression (guided)	06_logistic_poisson.R	2.5 to 2.6	20 min
C. Clustering (guided)	07_clustering.R	2.7	20 min
Final exercise	08_final_exercise.R	Final exercise	25 min
Review and close			5 min

- Extensions E2.1 to E2.4 are for those who finish early
- Keep the cheat sheet and decision diagrams open while you work

Reporting checklist

- State the question, design and number of observations used (and why any were excluded)
- Describe the data before modelling
- Report effect sizes with 95% CIs, in meaningful units
- Say which variables were adjusted for, and why
- Report checks of assumptions
- Separate statistical significance from practical importance
- Do not claim causation unless the design supports it



Copyright © Neudata, 2026

All property rights and copyright are reserved.

This presentation contains proprietary data, information, and materials developed by Neudata for dedicated use only and may not be communicated, copied, reproduced, distributed, published, or cited, in whole or in part, without the prior written consent of Neudata. Where explicit written permission has been granted, appropriate attribution must be included, followed by “by courtesy of Neudata”.

Any unauthorized use or infringement of these materials may give rise to legal action and claims for damages, without prejudice to any other rights of Neudata, including rights relating to patents, trademarks, or other forms of intellectual property protection.